

①

Vector Space

1. Definition

2. Problem

Subspace: Let V be a vector space over the field F . A subset W of a vector space $V(F)$ is called a subspace of V if W itself a vector space under the addition and scalar multiplication defined on V .

Example: ① Let $V = \mathbb{R}^2$. Then the subspace of $\mathbb{R}^2$ are

① $\{0\} = \{(0,0)\}$ ② Lines through the origin

② Let $V = \mathbb{R}^3$. Then the subspace of $\mathbb{R}^3$ are

① $\{0\} = \{(0,0,0)\}$ ② Lines through the origin

③ Planes through the origin.

③ Consider the vector space $V = \mathbb{R}^2 = \{(a,b) : a,b \in \mathbb{R}\}$. Then $W_1 = \{(a,0) : a \in \mathbb{R}\}$, $W_2 = \{(0,b) : b \in \mathbb{R}\}$ are subspaces of $V = \mathbb{R}^2$.

④ $W_1 = \{(a,b,0) : a,b \in \mathbb{R}\}$ and $W_2 = \{(0,b,c) : b,c \in \mathbb{R}\}$ are subspaces of $V = \mathbb{R}^3 = \{(a,b,c) : a,b,c \in \mathbb{R}\}$

* $W = \{(x,y) : x \geq 0, y \geq 0\}$ is not a subspace of $V = \mathbb{R}^2$ as it is not closed under scalar multiplication.

For example, $\underline{v} = (1,1) \in W$ but it's negative

$(-1)\underline{v} = (-1,-1) \notin W$.

(2)

Thm: Let V be a vector space over the field F . Then a subset W of V is a subspace of V if and only if

(a) $\underline{0} \in W$

(b) W is closed under addition i.e., $\underline{u}, \underline{v} \in W \Rightarrow \underline{u} + \underline{v} \in W$

(c) W is closed under scalar multiplication i.e., for all $\alpha \in F$ and for all $\underline{u} \in W$, $\alpha \underline{u} \in W$

Problem: Test whether the following subsets are subspace of $\mathbb{R}^3$ or not.

(i) $T = \{(a, b, c) \in \mathbb{R}^3 : a^2 + b^2 + c^2 \geq 1, a, b, c \in \mathbb{R}\}$

(ii) $S_1 = \{(a, b, c) \in \mathbb{R}^3 : a + c = -b, a, b, c \in \mathbb{R}\}$

(iii) $S_2 = \{(a, b, c) \in \mathbb{R}^3 : 3a + 2b - c = 0, a, b, c \in \mathbb{R}\}$

(iv) $T_1 = \{(a, b, c) : a^2 + b^2 + c^2 \leq 1, a, b, c \in \mathbb{R}\}$

(v) $T_2 = \{(a, b, 1) : a, b \in \mathbb{R}\}$

(vi) $T_3 = \{(a, b, c) : a \geq 0\}$

Solⁿ: (i) Given $T = \{(a, b, c) \in \mathbb{R}^3 : a^2 + b^2 + c^2 \geq 1\}$
 Hence $(0, 0, 0) \notin T$ as $0^2 + 0^2 + 0^2 = 0 < 1$.
 $\therefore T$ is not a subspace of $\mathbb{R}^3$.

(ii) Given $S_1 = \{(a, b, c) \in \mathbb{R}^3 : a + c = -b; a, b, c \in \mathbb{R}\}$
 Hence $(0, 0, 0) \in S_1$ as $0 + 0 = 0 = -0$.

Let $\underline{u}, \underline{v} \in S_1$ where $\underline{u} = (a, b, c)$ and $\underline{v} = (x, y, z)$
 Then $a + c = -b$ and $x + z = -y$ — (i)

Now $\underline{u} + \underline{v} = (a, b, c) + (x, y, z)$
 $= (a+x, b+y, c+z)$

$a+x + c+z = -b-y$ — (ii)
 $= -(b+y)$ — (iii)

(3)

Again let $\alpha \in \mathbb{R}$, then $\alpha \underline{u} = \alpha(a, b, c) = (\alpha a, \alpha b, \alpha c)$

$$\text{Now } \alpha a + \alpha c = \alpha(a+c) = -\alpha b \quad [\text{by } (*)]$$

$$\therefore \alpha \underline{u} \in S.$$

Thus S is a subspace of $\mathbb{R}^3$.

$$\textcircled{\text{iii}} \quad S_2 = \{(a, b, c) : 3a + 2b - c = 0\}$$

$$\textcircled{i} \quad \underline{0} = (0, 0, 0) \in S_2 \text{ as } 3 \cdot 0 + 2 \cdot 0 - 0 = 0.$$

$$\textcircled{ii} \quad \underline{u}, \underline{v} \in S_2, \quad \underline{u} = (a, b, c), \quad \underline{v} = (x, y, z)$$
$$3a + 2b - c = 0 \text{ and } 3x + 2y - z = 0$$

$$\underline{u} + \underline{v} = (a+x, b+y, c+z)$$

$$3(a+x) + 2(b+y) - (c+z)$$

$$= (3a + 2b - c) + (3x + 2y - z)$$

$$= 0 + 0$$

$$= 0$$

$$\therefore \underline{u} + \underline{v} \in S_2$$

$$\textcircled{\text{iii}} \quad \alpha \underline{u} = \alpha(a, b, c)$$

$$\text{Now } 3\alpha a + 2\alpha b - \alpha c$$

$$= \alpha(3a + 2b - c)$$

$$= 0$$

$$\therefore \alpha \underline{u} \in S_2$$

$\therefore S_2$ is a subspace of $\mathbb{R}^3$.

$$\textcircled{\text{iv}} \quad T_1 = \{(a, b, c) : a^2 + b^2 + c^2 \leq 1\}$$

$$\textcircled{i} \quad \underline{0} \in T_1 \text{ as } \dots$$

Let $\underline{u} = (1, 0, 0) \in T_1$ and $\alpha = 2 \in \mathbb{R}$, then

$$\alpha \underline{u} = (2, 0, 0) \notin T_1 \text{ as } 2^2 + 0 + 0 = 4 > 1.$$

(4)

* Show that $S = \{ (a, b, c, d) \in \mathbb{R}^4 : 2a - 3b + 5c - d = 0 \}$
is a subspace of $\mathbb{R}^4$.

(5)

Linear Independence and Linear Dependence

Linearly Independent and dependent set:

Let $S = \{ \underline{v}_1, \underline{v}_2, \dots, \underline{v}_n \}$ be a non empty set of vectors, then the vector equation $\alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \dots + \alpha_n \underline{v}_n = 0$ where $\alpha_i \in F$, has at least one solⁿ, namely

$$\alpha_1 = \alpha_2 = \dots = \alpha_n = 0$$

If this is the only solⁿ, then S is called a linearly Independent ~~set~~ ^{set}. If there are other solⁿs, then S is called linearly dependent set.

Ex-1: Linearly dependent set: Let $\underline{v}_1 = (2, -1, 0, 3)$

$\underline{v}_2 = (1, 2, 5, -1)$ and $\underline{v}_3 = (7, -1, 5, 8)$ then the set of vectors $S = \{ \underline{v}_1, \underline{v}_2, \underline{v}_3 \}$ is linearly dependent as $3\underline{v}_1 + \underline{v}_2 - \underline{v}_3 = 0$

Ex-2: Linearly Independent set:

$\underline{v}_1 = (1, 0, 0)$, $\underline{v}_2 = (0, 1, 0)$, $\underline{v}_3 = (0, 0, 1)$, then

$$\alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3 = 0$$

$$\Rightarrow \alpha_1(1, 0, 0) + \alpha_2(0, 1, 0) + \alpha_3(0, 0, 1) = (0, 0, 0)$$

$$\Rightarrow (\alpha_1, 0, 0) + (0, \alpha_2, 0) + (0, 0, \alpha_3) = (0, 0, 0)$$

$$\Rightarrow (\alpha_1 + 0 + 0, 0 + \alpha_2 + 0, 0 + 0 + \alpha_3) = (0, 0, 0)$$

$$\Rightarrow (\alpha_1, \alpha_2, \alpha_3) = (0, 0, 0)$$

$\therefore \alpha_1 = \alpha_2 = \alpha_3 = 0$, So $S = \{ \underline{v}_1, \underline{v}_2, \underline{v}_3 \}$ is linearly independent.

(6)

* Show that the vectors $\underline{v}_1 = (1, 1, -1)$, $\underline{v}_2 = (2, 1, 0)$ and $\underline{v}_3 = (-1, 1, 2)$ are linearly independent.

Solⁿ: Let $\alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3 = \underline{0}$

$$\Rightarrow (\alpha_1 + 2\alpha_2 - \alpha_3, \alpha_1 + \alpha_2 + \alpha_3, -\alpha_1 + \alpha_3) = (0, 0, 0)$$

Equating corresponding components and forming the linear system, we get

$$\alpha_1 + 2\alpha_2 - \alpha_3 = 0$$

$$\alpha_1 + \alpha_2 + \alpha_3 = 0$$

$$-\alpha_1 + \alpha_3 = 0$$

Let us solve the system. The augmented matrix of the system is

$$\left(\begin{array}{ccc|c} 1 & 2 & -1 & 0 \\ 1 & 1 & 1 & 0 \\ -1 & 0 & 1 & 0 \end{array} \right)$$

$$\begin{array}{l} \xrightarrow[R_3 \rightarrow R_3 + R_1]{R_2 \rightarrow R_2 - R_1} \quad \left| \quad \xrightarrow{R_2 \rightarrow (-1)R_2} \quad \left| \quad \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \quad \left| \quad \xrightarrow{R_3 \rightarrow R_3/4} \right. \right. \end{array}$$

After back substitution, ~~we~~ $\alpha_1 = \alpha_2 = \alpha_3 = 0$

[H.W.] Determine whether or not the following set of vectors/vectors in $\mathbb{R}^3$ are linearly dependent or independent.

① $\underline{v}_1 = (1, -2, -1)$, $\underline{v}_2 = (1, -4, 3)$ and $\underline{v}_3 = (0, 2, 4)$

② $\underline{v}_1 = (2, -1, 4)$, $\underline{v}_2 = (3, 6, 2)$ and $\underline{v}_3 = (2, 10, -4)$

⑦

③ $\underline{v}_1 = (1, -2, 3)$, $\underline{v}_2 = (5, 6, -1)$ and $\underline{v}_3 = (3, 2, 1)$
Linearly dependent

Hints:
$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ -2 & 6 & 2 & 0 \\ 3 & -1 & 1 & 0 \end{array} \right)$$

$$\left(\begin{array}{ccc|c} 1 & 5 & 3 & 0 \\ 0 & 1 & \frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{array} \right)$$

④ $\underline{v}_1 = (2, -1, 0, 3)$, $\underline{v}_2 = (1, 2, 5, -1)$ and
 $\underline{v}_3 = (7, -1, 5, 8)$
(Linearly dependent)

⑤ $\{(1, 1, 1), (0, 0, 0), (1, 1, 0)\}$ (Linearly dependent)

* Linear Combination: A vector $\underline{u}$ is called a linear combination of the vectors $\underline{v}_1, \underline{v}_2, \dots, \underline{v}_n$ if it can be expressed in the form

$$\underline{u} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \dots + \alpha_n \underline{v}_n; \quad \alpha_1, \alpha_2, \dots, \alpha_n \text{ are scalars.}$$

Problem: Show that $\underline{v} = (5, -2, 1)$ is a linear combination of the vectors $\underline{v}_1 = (-1, 2, 0)$, $\underline{v}_2 = (3, 1, 2)$ and $\underline{v}_3 = (0, 1, -1)$.

Solⁿ: Let $\underline{v} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3$

$$\Rightarrow (5, -2, 1) = \alpha_1 (-1, 2, 0) + \alpha_2 (3, 1, 2) + \alpha_3 (0, 1, -1)$$

$$\Rightarrow (5, -2, 1) = (-\alpha_1 + 3\alpha_2, 2\alpha_1 + \alpha_2 + \alpha_3, 2\alpha_2 - \alpha_3)$$

(8)

This gives

$$-\alpha_1 + 3\alpha_2 = 5$$

$$2\alpha_1 + \alpha_2 + \alpha_3 = -2$$

$$2\alpha_2 - \alpha_3 = 1$$

Augmented matrix:

$$\left(\begin{array}{ccc|c} -1 & 3 & 0 & 5 \\ 2 & 1 & 1 & -2 \\ 0 & 2 & -1 & 1 \end{array} \right)$$

$$\xrightarrow{R_2 \rightarrow R_2 + 2R_1} \left(\begin{array}{ccc|c} -1 & 3 & 0 & 5 \\ 0 & 7 & 1 & 8 \\ 0 & 2 & -1 & 1 \end{array} \right)$$

$$\xrightarrow{R_3 \rightarrow 7R_3 + (-2)R_2} \left(\begin{array}{ccc|c} -1 & 3 & 0 & 5 \\ 0 & 7 & 1 & 8 \\ 0 & 0 & -9 & -9 \end{array} \right)$$

$$\therefore \alpha_1 = -2, \alpha_2 = 1, \alpha_3 = 1$$

$$\therefore (5, -2, 1) = -2(-1, 2, 0) + 1(3, 1, 2) + 1(0, 1, -1)$$

i.e., $\underline{v}$ is a linear combination of $\underline{v}_1, \underline{v}_2, \underline{v}_3$.

H.W. (10) Determine whether the vector $\underline{v}$ is a linear combination of $\underline{v}_1, \underline{v}_2$ and $\underline{v}_3$

(i) $\underline{v} = (5, 9, 5), \underline{v}_1 = (2, 1, 4), \underline{v}_2 = (1, -1, 3), \underline{v}_3 = (3, 2, 5)$

Ans: $\underline{v} = 3\underline{v}_1 - 4\underline{v}_2 + \underline{v}_3$

(ii) $\underline{v} = (2, -5, 3), \underline{v}_1 = (1, -3, 2), \underline{v}_2 = (2, -4, -1), \underline{v}_3 = (1, -5, 7)$

Since the system is inconsistent, so no such scalars $\alpha_1, \alpha_2, \alpha_3$ exist s.t. $\underline{v} = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3$
~~where~~ $\underline{v}$ is not a linear combination of $\underline{v}_1, \underline{v}_2, \underline{v}_3$.

(9)

Linear Span or Generator:

Let $S = \{\underline{v}_1, \underline{v}_2, \dots, \underline{v}_n\}$ be a set of vectors in a vector space $V(F)$. The set S spans V or V is spanned or generated by S , if every vector in V is a linear combination of the vectors in S .

Problem-1: Show that $\{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ is a generator of a vector space $\mathbb{R}^3$.

or Determine whether $\{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ span or generate the vector space $\mathbb{R}^3$.

Solⁿ: We must determine whether an arbitrary vector $(a, b, c) \in \mathbb{R}^3$ can be expressed as a linear combination, $(a, b, c) = \alpha_1 \underline{v}_1 + \alpha_2 \underline{v}_2 + \alpha_3 \underline{v}_3$, $\alpha_1, \alpha_2, \alpha_3 \in \mathbb{R}$ of the vectors $\underline{v}_1, \underline{v}_2$ and $\underline{v}_3$. ①

~~Let~~

$$\textcircled{1} \Rightarrow \alpha_1(1, 2, 1) + \alpha_2(2, 1, 0) + \alpha_3(1, -1, 2) = (a, b, c)$$

$$\Rightarrow (\alpha_1 + 2\alpha_2 + \alpha_3, 2\alpha_1 + \alpha_2 - \alpha_3, \alpha_1 + 2\alpha_3) = (a, b, c)$$

; which gives the following system of linear eqns.

$$\left. \begin{array}{l} \alpha_1 + 2\alpha_2 + \alpha_3 = a \\ 2\alpha_1 + \alpha_2 - \alpha_3 = b \\ \alpha_1 + 2\alpha_3 = c \end{array} \right\} \text{--- } \textcircled{2}$$

Now the determinant of the co-efficient matrix of the system of linear eqns $\textcircled{2}$ is

$$D = \begin{vmatrix} 1 & 2 & 1 \\ 2 & 1 & -1 \\ 1 & 0 & 2 \end{vmatrix} = 1(2+0) - 2(4+1) + 1(0-1) = -9 \neq 0$$

Since $D \neq 0$, so solⁿ of $\textcircled{2}$ exists. Hence $(a, b, c) \in \mathbb{R}^3$ can be expressed as a linear combination of the vectors $(1, 2, 1), (2, 1, 0)$ and $(1, -1, 2)$.

(10)

This implies that $\{(1, 2, 1), (2, 1, 0), (1, -1, 2)\}$ is a generator of a vector space $\mathbb{R}^3$.
or $(1, 2, 1), (2, 1, 0)$ and $(1, -1, 2)$ generate (span) $\mathbb{R}^3$.

H.W. Determine whether the following vectors span or generate the vector space $\mathbb{R}^3$.

(i) $\underline{v}_1 = (1, 1, 2), \underline{v}_2 = (1, 0, 1)$ and $\underline{v}_3 = (2, 1, 3)$

Ans: ~~do~~ do not span.

(ii) $\underline{u} = (1, 2, 3), \underline{v} = (0, 1, 2)$ and $\underline{w} = (0, 0, 1)$

(iii) $\underline{u}_1 = (1, -2, 1), \underline{u}_2 = (2, -1, 1)$ and $\underline{u}_3 = (2, 3, 1)$

Basis and Dimension

Basis: A finite subset S of a vector space V over F is called a basis of V if

- (i) S is linearly independent
- (ii) S spans V

Problem-1: Show that $S = \{(1, 2), (2, 1)\} \subseteq \mathbb{R}^2$ is a basis of $V = \mathbb{R}^2$.

Solⁿ: To show this we need to check:

- (i) S is linearly independent
- (ii) S spans $\mathbb{R}^2$.

(i) Let $\alpha_1(1, 2) + \alpha_2(2, 1) = (0, 0)$ for some $\alpha_1, \alpha_2 \in \mathbb{R}$

$$\Rightarrow (\alpha_1 + 2\alpha_2, 2\alpha_1 + \alpha_2) = (0, 0)$$

$$\Rightarrow \begin{aligned} \alpha_1 + 2\alpha_2 &= 0 \\ 2\alpha_1 + \alpha_2 &= 0 \end{aligned}$$

(11)

let us solve this system. Its augmented matrix is

$$\begin{pmatrix} 1 & 2 & | & 0 \\ 2 & 1 & | & 0 \end{pmatrix}$$

$$\xrightarrow{R_2 \rightarrow R_2 - 2R_1} \begin{pmatrix} 1 & 2 & | & 0 \\ 0 & -3 & | & 0 \end{pmatrix}$$

$$R_3 \rightarrow (-\frac{1}{3})R_3 \begin{pmatrix} 1 & 2 & | & 0 \\ 0 & 1 & | & 0 \end{pmatrix}$$

$$\Rightarrow \alpha_1 + 2\alpha_2 = 0$$

$$\alpha_2 = 0$$

$$\therefore \alpha_1 = 0, \alpha_2 = 0$$

This implies that S is linearly independent in $\mathbb{R}^2$.

⑪ For any $(a, b) \in \mathbb{R}^2$, let $(a, b) = \alpha_1(1, 2) + \alpha_2(2, 1)$, $\alpha_1, \alpha_2 \in \mathbb{R}$

$$\Rightarrow \begin{cases} \alpha_1 + 2\alpha_2 = a \\ 2\alpha_1 + \alpha_2 = b \end{cases} \quad \text{--- ①}$$

The determinant of the coefficient matrix of ① is

$$D = \begin{vmatrix} 1 & 2 \\ 2 & 1 \end{vmatrix} = 1 - 4 = -3 \neq 0$$

Hence S spans $\mathbb{R}^2$.

$\therefore S$ is a basis of $\mathbb{R}^2$.

H.W. Which of the following sets of vectors are basis for $\mathbb{R}^3$?

① $\underline{v}_1 = (1, 2, 1), \underline{v}_2 = (2, 9, 0), \underline{v}_3 = (3, 3, 4)$

② $\underline{v}_1 = (4, 2, 1), \underline{v}_2 = (2, 6, -5), \underline{v}_3 = (1, -2, 3)$

③ $(1, 0, 0), (2, 2, 0), (3, 3, 3)$

Linear Transformation

Linear Transformation or mapping and Linear Operator:

Let $T: V \rightarrow W$ be a function from a vector space V into a vector space W . If T satisfies the following conditions

$$\textcircled{i} T(a+b) = T(a) + T(b); \forall a, b \in V$$

$$\textcircled{ii} T(\alpha a) = \alpha T(a); \forall a \in V, \alpha \in \mathbb{R}$$

Then T is called a linear transformation or mapping.

In the special cases where $V=W$, the linear transformation or mapping $T: V \rightarrow V$ is called a linear operator on V .

* Show that the transformation $T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$,
 $T(x, y, z) = (z, x+y)$ is linear.

Solⁿ: Let $\underline{u}, \underline{v} \in \mathbb{R}^3$, where $\underline{u} = (x_1, y_1, z_1)$ and $\underline{v} = (x_2, y_2, z_2)$

$$\text{Then } T(\underline{u}) = T(x_1, y_1, z_1) = (z_1, x_1 + y_1)$$

$$\text{and } T(\underline{v}) = T(x_2, y_2, z_2) = (z_2, x_2 + y_2)$$

$$\underline{u} + \underline{v} = (x_1, y_1, z_1) + (x_2, y_2, z_2)$$

$$= (x_1 + x_2, y_1 + y_2, z_1 + z_2)$$

$$\text{and } T(\underline{u} + \underline{v}) = (z_1 + z_2, x_1 + x_2 + y_1 + y_2)$$

$$= (z_1, x_1 + y_1) + (z_2, x_2 + y_2)$$

$$= T(\underline{u}) + T(\underline{v})$$

Also, if $\alpha \in \mathbb{R}$, $\alpha \underline{u} = (\alpha x_1, \alpha y_1, \alpha z_1)$ such that

$$T(\alpha \underline{u}) = T(\alpha x_1, \alpha y_1, \alpha z_1)$$

$$= (\alpha z_1, \alpha x_1 + \alpha y_1)$$

$$= \alpha (z_1, x_1 + y_1)$$

$$= \alpha T(\underline{u})$$

Thus the transformation T is linear.

(13)

* let $T: \mathbb{R} \rightarrow \mathbb{R}$ be the transformation defined by $T(x) = 2x + 1$. Check whether this is a linear transformation.

Sup: $x, y \in \mathbb{R}$

$$T(x+y) = 2(x+y) + 1 \\ = 2x + 2y + 1$$

$$T(x) + T(y) = (2x+1) + (2y+1) \\ = 2x + 2y + 2$$

$$\therefore T(x+y) \neq T(x) + T(y)$$

* let $T: \mathbb{R}^2 \rightarrow \mathbb{R}^3$ be the transformation defined by $T(\underline{u}) = (x, x+y, x-y)$ where $\underline{u} = (x, y)$ is a vector in $\mathbb{R}^2$. Then T is a linear transformation.

Sup: $\underline{u} = (x_1, y_1), \underline{v} = (x_2, y_2)$

$$\underline{u} + \underline{v} = (x_1 + x_2, y_1 + y_2)$$

$$T(\underline{u} + \underline{v}) = (x_1 + x_2, (x_1 + x_2) + (y_1 + y_2), (x_1 + x_2) - (y_1 + y_2)) \\ = (x_1, x_1 + y_1, x_1 - y_1) + (x_2, x_2 + y_2, x_2 - y_2) \\ = T(\underline{u}) + T(\underline{v})$$

$$\alpha \in \mathbb{R}, \alpha \underline{u} = (\alpha x_1, \alpha y_1)$$

$$T(\alpha \underline{u}) = T(\alpha x_1, \alpha y_1) \\ = (\alpha x_1, \alpha x_1 + \alpha y_1, \alpha x_1 - \alpha y_1) \\ = \alpha (x_1, x_1 + y_1, x_1 - y_1) \\ = \alpha T(\underline{u})$$

(14)

$$* T: \mathbb{R}^3 \rightarrow \mathbb{R}^2, T(x, y, z) = (x+1, y+z)$$

show T is not linear.

$$\begin{aligned} \underline{u} + \underline{v} &= (x_1, y_1, z_1) + (x_2, y_2, z_2) \\ &= (x_1+x_2, y_1+y_2, z_1+z_2) \end{aligned}$$

$$T(\underline{u} + \underline{v}) = (x_1+x_2+1, y_1+y_2+z_1+z_2)$$

$$\begin{aligned} \# T(\underline{u}) + T(\underline{v}) &= (x_1+1, y_1+z_1) + (x_2+1, y_2+z_2) \\ &= (x_1+x_2+2, y_1+y_2+z_1+z_2) \end{aligned}$$

□

$$\boxed{\text{H.W}} \quad \textcircled{I} T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$$

$$T(x, y) = (x-y, x+y)$$

$$\textcircled{II} T: \mathbb{R}^3 \rightarrow \mathbb{R}^2$$

$$T(x, y, z) = (3x-2y+z, x-3y-2z)$$

(15)

Eigenvalues and Eigenvectors

If A is an $n \times n$ matrix, then a non zero vector V in $\mathbb{R}^n$ is called an eigenvector of A if AV is a scalar multiple of V i.e; $AV = \lambda V$ for some scalar λ . The scalar λ is called an eigenvalue of A and V is said to be an eigenvector of A corresponding to λ .

Example: The vector $V = \begin{pmatrix} 2 \\ 1 \end{pmatrix}$ is an eigenvector of the matrix $A = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix}$ corresponding to the eigenvalue $\lambda = 2$. Since $AV = \begin{pmatrix} 1 & 2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 \\ 1 \end{pmatrix} = \begin{pmatrix} 4 \\ 2 \end{pmatrix} = 2 \begin{pmatrix} 2 \\ 1 \end{pmatrix} = 2V$.

Characteristic Matrix, Polynomial and Equⁿ:

If A is an $n \times n$ matrix, then (i) $A - \lambda I_n$ is called the characteristic matrix of A , where I_n is an $n \times n$ Identity matrix.

(ii) the determinant of $A - \lambda I_n$ i.e; $|A - \lambda I_n|$ is called the characteristic polynomial of A

(iii) the equⁿ $|A - \lambda I_n| = 0$ is called the characteristic equⁿ of A

Example: Let $A = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix}$. The characteristic matrix

$$\text{of } A \text{ is } A - \lambda I_2 = \begin{pmatrix} 1 & 1 \\ 2 & 3 \end{pmatrix} - \lambda \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \\ = \begin{pmatrix} 1-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix}$$

$$\text{Characteristic polynomial is } |A - \lambda I_2| = \begin{vmatrix} 1-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} \\ = \lambda^2 - 4\lambda + 1$$

$$\text{Characteristic equⁿ is } \lambda^2 - 4\lambda + 1 = 0$$

(16)

* Find the eigenvalues and associated eigenvectors of the following matrices.

① $\begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$ ② $\begin{pmatrix} 2 & 3 \\ 1 & 4 \end{pmatrix}$ ③ $\begin{pmatrix} 2 & 3 \\ 2 & 1 \end{pmatrix}$ ④ $\begin{pmatrix} 2 & 2 \\ 1 & 3 \end{pmatrix}$

① Let $A = \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix}$. The characteristic eqn of A is

$$|A - \lambda I_2| = 0$$

$$\Rightarrow \left| \begin{pmatrix} 2 & 1 \\ 2 & 3 \end{pmatrix} - \begin{pmatrix} \lambda & 0 \\ 0 & \lambda \end{pmatrix} \right| = 0$$

$$\Rightarrow \begin{vmatrix} 2-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$\Rightarrow \lambda^2 - 5\lambda + 4 = 0$$

$$\Rightarrow (\lambda - 4)(\lambda - 1) = 0$$

$\therefore \lambda = 4, 1$; which are the eigenvalues of A.

Let $\underline{v} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} \neq \underline{0}$ be the eigenvector of A corresponding to $\lambda = 4$ and $\lambda = 1$.

Then by defⁿ of eigenvector, we get

$$(A - \lambda I_2) \underline{v} = \underline{0}$$

$$\Rightarrow \begin{pmatrix} 2-\lambda & 1 \\ 2 & 3-\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \quad \text{--- ①}$$

Putting $\lambda = 4$ in ①, we find

$$\begin{pmatrix} 2-4 & 1 \\ 2 & 3-4 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2 & 1 \\ 2 & -1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -2x_1 + x_2 \\ 2x_1 - x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

(17)

Which gives the following system of linear eqns:

$$-2x_1 + x_2 = 0$$

$$2x_1 - x_2 = 0$$

$$\xrightarrow{4 \rightarrow (-1)H} \begin{array}{l} 2x_1 - x_2 = 0 \\ 2x_1 - x_2 = 0 \end{array}$$

$$\longrightarrow 2x_1 - x_2 = 0$$

Here x_2 is a free variable. Let $x_2 = a$, where $a \in \mathbb{R}$ but $a \neq 0$.

$$\text{Then } x_1 = \frac{a}{2} \therefore \underline{v} = \begin{pmatrix} \frac{a}{2} \\ a \end{pmatrix}$$

Taking $a=2$, we get $\underline{v} = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$; which is a eigenvector of the matrix A corresponding to $\lambda = 4$.

Now putting $\lambda = 1$ in (1), we get

$$\therefore \therefore$$

$$x_1 + x_2 = 0$$

$$\text{Let } x_2 = a, \text{ then } x_1 = -a \therefore \underline{v} = \begin{pmatrix} -a \\ a \end{pmatrix}$$

$$\text{For } a=1, \underline{v} = \begin{pmatrix} -1 \\ 1 \end{pmatrix}; \text{ which } \dots \text{ for } \lambda=1$$